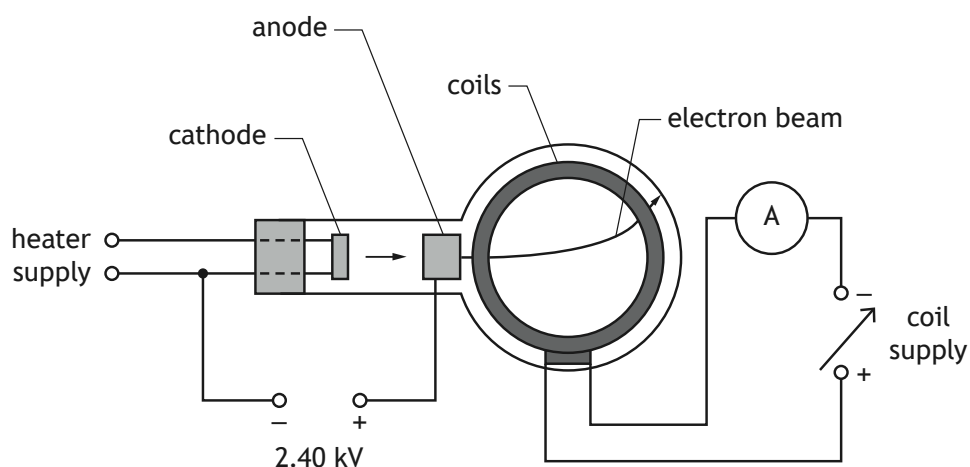


6. A teacher uses the apparatus shown to demonstrate the deflection of a beam of electrons by a magnetic field.



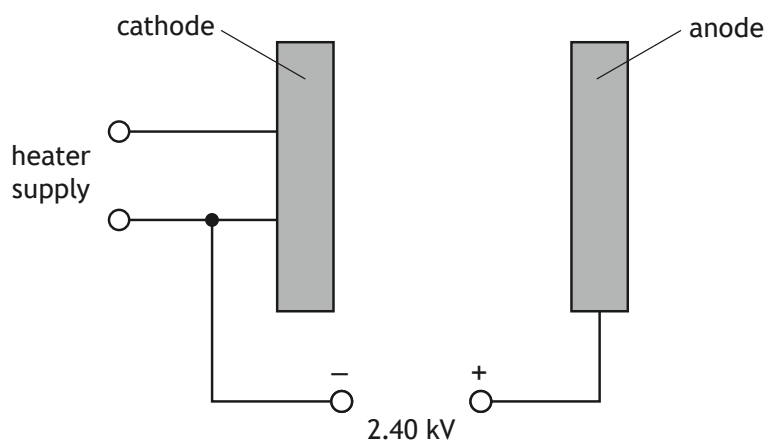
- (a) The heater supply is used to heat the cathode, which causes electrons to be released from its surface.

The electrons are accelerated from rest towards the anode by a potential difference of 2.40 kV.

- (i) Complete the diagram below, to show the electric field pattern between the cathode and anode.

1

(An additional diagram, if required, can be found on *page 53*.)



6. (a) (continued)

- (ii) The gain in kinetic energy for each electron moving from the cathode to the anode is  $3.84 \times 10^{-16}$  J.

Calculate the maximum speed of an electron as it reaches the anode.

3

*Space for working and answer*

- (iii) The cathode and anode are separated by a distance of 45.0 mm.

Calculate the maximum acceleration of an electron between the cathode and the anode.

3

*Space for working and answer*

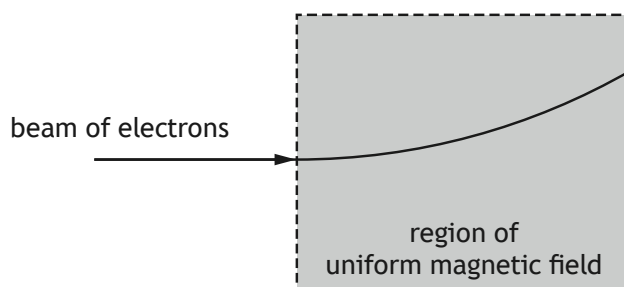
[Turn over



\* X 8 5 7 7 6 0 1 2 1 \*

6. (continued)

- (b) A small hole in the anode allows a narrow beam of electrons to be produced. The beam of electrons passes into a uniform magnetic field between the coils. The diagram shows the path taken by the electrons in the uniform magnetic field.



Determine the direction of the uniform magnetic field.

1